Satellite-Anchored Pre-Earnings Trading in Permian E&P:

A Multi-Agent Framework with a Deterministic Numerical Core

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Abstract

We design and test a five-agent system that combines real Sentinel-1 SAR drilling-pad activity, point-in-time IBES analyst consensus, and LLM-driven news verification to produce pre-earnings long trades for ten Permian-basin E&P firms over Q1 2019 – Q4 2024 (200 firm-quarter cells). The system separates a *deterministic numerical core* (consensus, revenue forecast, divergence threshold, conviction-to-size lookup) from an *LLM-driven qualitative shell* (outlook synthesis, news verification, Bull/Bear/Arbiter board). Numbers that drive trades are produced in code; LLMs produce annotations.

All inputs are real public data: Sentinel-1 RTC backscatter, FracFocus completions, EIA WTI and rig count, IBES Detail-History, GDELT, CRSP+Yahoo. The system produced **51 long trades** across 2019–2024: **27W / 24L (52.9%)**, **+0.18% mean net return per trade**, **+0.88% median**, **+\$6,634 on \$1M capital** (+0.66%). Firm-clustered bootstrap fails to reject H0 = 50% (**p = 0.328**, 95% CI **[39.5%, 65.1%]**); 95% CI on mean return **[-1.83%, +1.53%]** straddles zero. Two deterministic ablations produce zero entries: $\alpha = 0$ and forcing Agent 1 to all-idle.

The aggregate is not driven by a single dominant trade. Top contributors: SM 2021-Q3 (+\$24,453), OVV 2023-Q2 (+\$15,504); worst MTDR 2020-Q3 (−\$21,509). The most consequential structural pattern is **2021 over-firing**: 16 long trades in 2021 (post-COVID drilling recovery), 13 clipped at `drilling_signal = +1.0`, pinning forecast divergence at exactly +10% and triggering `modest_beat`. The clip works as designed but collapses signal differentiation across the recovery cohort: the gate cannot distinguish a modest recovery from an explosive one. We disclose rather than tune post-hoc.

The contribution is methodological: a multi-agent design pattern with clearly separated LLM and deterministic roles, explicit ablations measuring component value, and a reproducibility manifest sufficient to replicate every trade. We do not claim profitability after realistic execution costs — the aggregate (+0.66% over 6 years, +0.18% mean per trade) is within the 30 bps round-trip cost band. We claim that separating deterministic from LLM roles, ablation-validating component contribution, and pre-registering tests is a more defensible foundation for agentic-AI trading than delegating numerical decisions to language models.

Keywords: multi-agent LLM, alternative data, satellite imagery, point-in-time consensus, ablation testing.

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2. Introduction

Trading around earnings is a classic battleground for information advantage. The two-week pre-announcement window is short enough that fundamental drift is small but long enough that a forecast a few percent better than consensus can be turned into directional return. This paper asks whether a multi-agent system combining satellite-derived drilling activity, point-in-time analyst consensus, and LLM-based news verification can produce such a forecast for U.S. Permian E&P firms — and whether the satellite signal contributes incrementally over consensus alone.

We choose Permian E&P deliberately. The basin is visible from SAR satellites; most U.S. small- and mid-cap operators derive a majority of revenue from it; revenue is mechanistically tied to observable inputs (active pads, completion intensity, WTI prices); and per-name analyst panels are thin enough that consensus is sometimes stale at quarter-end.

The system is **multi-agent**: Agent 1 (SAR pad activity, no LLM); Agent 2 (consensus-anchored deterministic forecast + LLM outlook); Agent 3 (point-in-time IBES consensus comparison + divergence class); Agent 4 (GDELT verification); Agent 5 (Bull/Bear/Arbiter, deterministic conviction-to-size lookup). **Numeric outputs driving trades are code, not LLMs**: Agent 2's forecast is the deterministic value, Agent 3's class is Python, Agent 5's size is a fixed function of conviction tier. LLMs do qualitative annotation, narrative synthesis, and adversarial debate.

This separation is the core methodological position. A common failure mode of agentic-AI trading is delegating a numerical decision to a language model whose calibration cannot be verified, then inferring the system "works" from ex-post P&L — statistically indistinguishable from a black box. Our system makes every trade traceable to explicit numerical inputs and each LLM agent's marginal value measurable by ablation.

We evaluate Strategy 1 across **Q1 2019 – Q4 2024** against nine alternative strategies on the 2024 sub-window. Baselines span ablations (Strategy 2, no-news), deterministic single-signal alt-data (S3 analyst-revision, S4 WTI momentum, S5 EIA-DPR rig count), and conventional cross-sectional or passive baselines (S6–S10). All ten share an identical rebalance schedule, 30-bps round-trip cost, and \$1M starting capital. Pre-registration follows Harvey & Liu (2014): the primary test is a one-sided 5%-level firm-clustered bootstrap of hit rate vs H0 = 50%; window, universe, and holding period were fixed before any results were observed.

Contributions: (1) a multi-agent design pattern with clearly separated LLM and deterministic roles plus ablations confirming the satellite signal's marginal value; (2) point-in-time IBES consensus reconstruction under a strict estimate-active-at-T rule with full reproducibility; (3) real Sentinel-1 RTC integration via Microsoft Planetary Computer.

§3 literature; §4 hypotheses; §5 data; §§6–8 architecture, methodology, portfolio construction; §§9–11 results and ablations; §12 limitations; §13 conclusion. Appendix contains the per-trade ledger and reproducibility material.

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3. Literature Review

Satellite alt-data in commodities. Katona et al. (2018) established satellite imagery as a financial signal via parking-lot fill rates predicting retail same-store-sales surprises. Kang & Stulz (2021) extended to industrial activity around earnings; Li et al. (2022) showed Cushing storage-tank fill rates correlate 0.7–0.8 with EIA releases; Glaeser, Olsen & Welch (2020) used port-loading data for trade-dependent earnings. The lesson we adopt: imagery's predictive content is highest when mapped through a structural model to a financial line item, not fed directly into return regressions.

For Permian drilling, the closest reference is Ben-David et al. (2021), who used L-band SAR to track active and idle pads in the Delaware sub-basin with 91% detection accuracy on newly active pads, calibrated against Texas Railroad Commission and New Mexico Oil Conservation Division permits. This anchors our change-detection thresholds (≥1.5 dB activation / ≥0.5 dB sustained).

Three broader priors we adopt: (i) signal decay — alt-data edges arbitrage away in 12–36 months once commercialised (Mukherjee, Panayotov & Shon 2021); (ii) conditional value — most valuable for idiosyncratic, hard-to-cover names (Zhu 2019); (iii) stale-consensus pickup — marginal alpha is largest when consensus is being revised but has not reached new equilibrium (Bradshaw et al. 2017). The Permian universe matches all three.

Multi-agent LLM systems in finance. The closest comparable is Yang et al. (2024) "FinAgent" — market-data/news/debate agents for equity portfolio construction with a fully LLM-driven pipeline (position sizes and confidence are language-model outputs). Our contribution is the deterministic-numerical-core architecture: numbers driving P&L are code, LLMs do qualitative annotation. Other systems — FinMem (Yu et al. 2024), FinRL-Trader (Liu et al. 2023), TradingGPT (Wang et al. 2024) — do not use a deterministic-core / LLM-shell decomposition with explicit ablation. Park et al. (2023) show multi-agent debate produces more rigorous reasoning than single-prompt baselines; our Bull/Bear/Arbiter board inherits from that line, with the Arbiter's output mechanically lifted to a position size.

Point-in-time fundamentals. Ljungqvist, Malloy & Marston (2009) document look-ahead biases from backfilled databases; Anderson & Akbas (2020) show anomaly returns inflate by 30–50% without point-in-time discipline. We follow Bradshaw et al. (2017): consensus at T is the median of estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`). To our knowledge this is the first multi-agent LLM trading system using strict point-in-time IBES reconstruction.

GDELT as a corroboration signal. GDELT 2.0 (Bonaparte 2017) is a low-cost news source; Rambaccussing & Kwiatkowski (2020) confirm GDELT sentiment correlates with volatility but is a noisy direction predictor at sub-month horizons. We use GDELT as a verification filter, not a primary signal.

Where this paper sits. Intersection of commodity satellite alt-data, multi-agent LLM orchestration with a deterministic numerical core, and strict point-in-time consensus reconstruction. To our knowledge no prior work has measured the marginal contribution of a satellite signal in a system that is otherwise a faithful consensus tracker — which the $\alpha = 0$ ablation tests.

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4. Hypotheses and Research Questions

We pre-register two primary hypotheses and three secondary research questions. Every test below was specified before any P&L was computed; the headline test is one-sided at the 5% level.

4.1 Primary hypotheses

H1 (signal-direction). Strategy 1's hit rate exceeds 50% over Q1 2019 – Q4 2024. Test: firm-clustered bootstrap with 1,000 iterations, null-centered under H0 = 50%, one-sided, reject at $p < 0.05$ (Harvey & Liu 2014). Firm-clustering accommodates within-firm dependence across repeated quarterly trades without parametric assumptions on the return distribution. The system's high-precision-low-recall design produces ~3.8 long trades per year, so n is small and statistical power is limited.

H2 (signal-marginal-value). Strategy 1's hit rate exceeds Strategy 3's (analyst-revision follower) by at least three percentage points. Test: difference of means with quarter-block bootstrap, one-sided at 5%. The 3-pp threshold matches the typical pre-arbitrage edge in Mukherjee et al. (2021). RQ2 (the $\alpha = 0$ ablation) provides a cleaner internal counterfactual.

4.2 Secondary research questions

RQ1 (SAR threshold sensitivity). Sweep the change-detection thresholds. Deferred; thresholds are anchored ex ante to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020), not tuned in-sample.

RQ2 (α = 0 ablation). When α is set to zero, Agent 2's forecast equals consensus and the satellite signal cannot influence the divergence class. RQ2 isolates the satellite's marginal contribution: an $\alpha = 0$ hit rate close to the $\alpha = 0.10$ rate would imply the edge comes from elsewhere.

RQ3 (no-satellite ablation). Force Agent 1 to emit all-idle classifications. RQ3 disables satellite information at the input rather than the weighting stage; it is the cleanest test of whether the LLM scaffolding has signal independent of the satellite input.

4.3 Out-of-scope claims

We do not claim a deployable trading product. The 51-trade sample is still small; the IBES `ANNDATS_ACT` proxy is documented; the holding-period assumption (T-14 entry, T+2 exit) ignores overnight gap risk and microstructure costs above 30 bps. The paper's claim is methodological: a multi-agent system with a deterministic numerical core can ingest real Sentinel-1 SAR end-to-end across six calendar years and produce gated trade decisions whose component contributions are measurable via mechanical ablations.

4.4 What would falsify the paper

Three results would falsify the central thesis. (i) H1 fails — Strategy 1's hit rate does not exceed 50%. (ii) H2 fails — Strategy 1 does not exceed Strategy 3 by the pre-registered 3-pp threshold. (iii) RQ2 / RQ3 produce a hit rate at least as high as the full-system rate, implying the satellite signal is not the source of edge. We pre-register these criteria and report against them in §§9–11 regardless of outcome.

5. Data and Investment Universe

5.1 Universe and trading window

Ten U.S. E&P firms selected on Permian exposure: FANG, EOG, DVN, CTRA, OXY, MTDR, PR, OVV, SM, CRGY. Nine derive ≥30% of trailing-12-month production from the Delaware or Midland sub-basins. CRGY is a known coverage exception (Uinta/Eagle Ford footprint, zero Permian FracFocus records — Agent 1 classifies its cells as idle). Global majors are excluded.

System is **long-only**; below-threshold divergence holds cash in BIL. Headline window **Q1 2019 – Q4 2024** (six full calendar years, 200 firm-quarter cells after corporate-action exclusions), covering pre-COVID, the 2020 collapse and rebound, the 2021–2022 spike, and the 2023–2024 range. Decision date `T = earnings_date − 14 days`; holding window `T-14` entry to `T+2` exit.

5.2 Data sources and point-in-time discipline

All sources free-public or via WRDS.

Equity fundamentals. IBES Detail-History (`tr_ibes`, `SAL`): active panel at T is estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`); consensus = median. Compustat `comp.fundq` for value/quality baselines lagged to most recent reported quarter as of T. CRSP daily total-return through 2024-12-31; yfinance Adj Close for 2025.

Permits. FracFocus bulk CSV filtered to Permian counties and the ten operators (or predecessors after merger): 12,376 completion records. Approximations: completion = JobStartDate, spud = JobStartDate − 60d, permit = spud − 90d; only filings ≤ T eligible. Same data anchors Sentinel-1 pad selection.

Satellite SAR. Sentinel-1 RTC VV from Microsoft Planetary Computer via STAC API + cloud-optimised GeoTIFF range-reads. Per cell: 25 representative pads (deduplicated, stratified across active / recently-completed / older); all VV scenes within 365 days ending at T. One STAC search per firm-quarter at union bbox, each scene opened once (~12 hours full-sweep wall-clock). Per-pad caches make re-runs free.

Other. EIA weekly WTI spot. EIA-DPR Permian rig counts (May–Dec 2024 carried-forward, documented). GDELT 2.0 DOC API for Agent 4 (prior earnings to T-14). Earnings dates use IBES `ANNDATS_ACT` as proxy for pre-T-14 provability (Wall Street Horizon not subscribed).

5.3 Coverage audit

A live IBES coverage audit precedes the backtest. Of 200 nominal cells, 186 are trade-eligible (≥3 analysts at T-14). The 14 exclusions correspond to corporate-action timing: CTRA enters Q3 2021 (Cabot–Cimarex merger), CRGY Q1 2022 (Dec 2021 IPO), PR Q3 2022 (Centennial–Colgate merger).

5.4 Reproducibility footprint

Every run writes `manifest.json` recording SHA256 of input CSVs, SAR-mode flag, change-detection thresholds, α, per-agent provider/model identifiers, prompt SHAs, Python version, platform. LLM-call cache is keyed on `(prompt_sha, input_sha, model_id, model_version, temperature)`. Full material in Appendix C–D.

6. Multi-Agent Architecture

Five specialist agents in a sequential pipeline followed by an adversarial debate. Each maps to a distinct role from institutional workflows — alt-data analyst, modelling analyst, sell-side consensus analyst, news analyst, portfolio committee. Agents communicate through pydantic-validated JSON, serving as contract, audit artifact, and unit of attribution.

6.1 Agent inventory

Agent 1 — GIS Detection (no LLM). Consumes real Sentinel-1 RTC VV backscatter at FracFocus pad coordinates and applies the change-detection rule (§7.2) to produce per-pad `newly_active` / `continuously_active` / `idle` classifications, aggregated per cell into three counts, `share_active`, and `relative_activity_delta` (normalised against a 30%-of-pads baseline). Normalisation isolates *change*, not operator size.

Agent 2 — Revenue Forecast (deterministic core + LLM outlook). The forecast `forecast = consensus × (1 + α × clip(drilling_signal, −1, +1))` with $\alpha = 0.10$ frozen ex ante is computed by Python — never the LLM. The LLM (gpt-4o-mini) reads the deterministic number and writes a qualitative outlook plus key drivers.

Agent 3 — Consensus Comparison. Pulls IBES consensus at T-14 and classifies divergence as one of `{strong_beat, modest_beat, in_line, modest_miss, strong_miss}` with a confidence rating. Class, percentage, and confidence are deterministic Python after the LLM call (text only); a schema validator rejects any output where the LLM's class disagrees with the rule.

Agent 4 — News Verification (LLM, gpt-4o-mini). Reads GDELT articles between prior earnings and T-14; a positive `gdelt_disclosed` finding downgrades conviction one tier.

Agent 5 — Investment Board. Bull and Bear (gpt-4o-mini) each produce a constrained `BoardMemberOpinion`. The Arbiter (gpt-5-mini) consumes both plus the upstream summary and emits the final decision plus a structured `upstream_agent_summary` used for attribution. Size is a fixed lookup from conviction tier.

6.2 Coordination

Sequential at the cell level: each `(ticker × fiscal_quarter_end)` runs Agents 1 → 5, with each agent's validated output as the next agent's input. No inter-cell communication.

Two coordination choices matter. First, a **hard divergence-class gate** at Agent 3 short-circuits the cell to `no_trade` whenever divergence is not in `{modest_beat, strong_beat}`; when it fires, Agents 4 and 5 do not run. Second, **conviction-to-size is a fixed lookup** (`high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`); the Arbiter chooses tier, code chooses size.

6.3 Why multi-agent

A single end-to-end LLM call would be simpler and likely produce comparable headline performance. The multi-agent design is justified on three grounds: (i) explicit role separation makes inference inspectable and supports per-trade attribution; (ii) Bull/Bear/Arbiter debate produces structured disagreement a single agent cannot represent, enabling debate value as a distinct ablation; (iii) the deterministic core in Agent 2 separates qualitative interpretation (LLMs' strength) from numerical magnitude (their weakness). Whether these benefits translate to measurable gains is itself an empirical question.

7. Methodology

7.1 The signal chain

From raw radar pixels at FracFocus pad coordinates, through a literature-anchored change-detection rule, through five sequential agents, to a long-only trade decision — anchored on point-in-time information available before each earnings.

7.2 Sentinel-1 SAR ingestion

Sentinel-1 RTC imagery (VV only) from Microsoft Planetary Computer via STAC API + cloud-optimised GeoTIFF range-reads. Per cell, 25 representative pads (deduplicated, stratified). ~500m × 500m AOI per pad, **one STAC search per firm-quarter** at union bbox, each scene opened once and clipped to all 25 pad windows from the in-memory raster.

Classification rule, with `cur_db` the target-quarter mean VV and `base_db` the trailing-4Q mean for the same pad:

- **`newly_active`** if `cur_db ≥ base_db + 1.5 dB` AND no FracFocus completion record before T;
- **`continuously_active`** if `cur_db ≥ base_db + 0.5 dB` AND a completion within prior 8 quarters;
- **`idle`** otherwise.

Thresholds (1.5 dB / 0.5 dB) anchored to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020), not tuned in-sample. Activity normalisation `relative_activity_delta = absolute_active − trailing_4Q_avg` mitigates size confounding.

7.3 Point-in-time discipline

FracFocus masked to `JobStartDate ≤ T`. Sentinel-1 scenes filtered to `datetime ≤ T`. IBES consensus is the median of estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`) — the REVDATS filter eliminates the survivorship leak in Anderson & Akbas (2020). GDELT filtered to publish dates ≤ T-14. EIA WTI uses prints available before T. Earnings dates use IBES `ANNDATS_ACT` as documented proxy.

7.4 Consensus-anchored revenue forecast

Agent 2's deterministic core overlays the IBES prior:

```
drilling_signal = relative_activity_delta / max(trailing_4Q_avg, 1) forecast_revenue
= consensus_revenue × (1 + α × clip(drilling_signal, −1, +1))
```

α frozen ex ante: **α = 0.10** for the headline, **α = 0.0** as no-satellite ablation. Reframes the question from "did our model beat consensus" (a claim we do not make) to "did the satellite signal justify a bounded, principled disagreement with consensus." Documented fallback model when IBES coverage is unavailable.

7.5 Trade-decision logic

Long entry requires Agent 3's `divergence_class` ∈ `{modest_beat (≥ +5%), strong_beat (≥ +15%)}`. Under $\alpha = 0.10$, max achievable divergence is +10%, so `modest_beat` is the operative signal. Agent 4's `gdelt_disclosed` downgrades conviction one tier. Lookup maps Arbiter tier to size (15% / 10% / 5% / 0%). Agent-3 gate short-circuits to `no_trade` otherwise (Agents 4 and 5 do not run).

Trades enter at the close on T-14 and exit on the second trading day strictly after earnings. Total-return-with-dividends; 30 bps round-trip cost; long-only with 15% per-cell cap.

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8. Portfolio Construction and Risk Management

Portfolio rules are deliberately rigid to prevent the multi-agent system from drifting into ad-hoc allocation. Sizing is a deterministic conviction-to-size lookup, so the Investment Board's only discretion is *which* trades to take, not *how much* capital to allocate.

Sizing. The Arbiter assigns each long candidate one of four conviction tiers — `high`, `medium`, `low`, `none`. The lookup is fixed at `high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`. The 15% per-name cap binds even when conviction would otherwise warrant more, preventing single-name concentration regardless of LLM reasoning quality.

Concurrency cap (documented, non-binding in this sample). The runner manifest records `max_names: 8` as a documented portfolio-construction parameter. In the empirical 2019–2024 sample the cap never binds — the maximum simultaneous longs in any single quarter is 2021-Q3 with 8 trades (every then-eligible firm), exactly at the cap. The inference pipeline does not enforce the cap; the headline P&L in §9 counts every long the system generated. We disclose this gap in §12.6 rather than enforce the cap retroactively.

Cash sleeve. Unallocated capital is held in BIL (SPDR Bloomberg 1–3 Month T-Bill ETF) rather than zero-yield cash. BIL is the system's only non-WRDS price input; it keeps the cash sleeve tradable, free, and total-return-comparable to equity legs.

Trade timing. All entries open at the close on T-14 (two weeks before earnings) and close on the second trading day after the earnings report (T+2). The holding window is short enough that we treat WTI level as approximately constant and long enough to capture the earnings-surprise reaction.

Costs. A flat 30 bps round-trip cost is applied to entry value. We do not separately model bid-ask spread, market impact, borrow, or earnings-event slippage; the long-only restriction makes these simplifications defensible.

Risk attribution. Because sizing is deterministic given conviction tier, realised return decomposes cleanly across the agent stack: name selection is the LLM-driven step (Investment Board plus upstream agents), magnitude is the deterministic step. The attribution analysis in §11 leverages this separation.

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9. Results: Headline

9.1 Aggregate metrics, 2019–2024

Across **Q1 2019 – Q4 2024 (200 cells)**, the system produced 51 long entries. Rebalance T = earnings_date – 14 (exit T+2 trading days), 30 bps round-trip, \$1M capital. Full ledger in Appendix A.

Table 9.1 — Aggregate metrics, 2019–2024:

Metric	Value
Cells evaluated	200
Long trades fired	**51**
Wins / Losses	**27W / 24L**
Hit rate	**52.9%**
Mean net return per trade	**+0.18%**
Median net return per trade	+0.88%
Total net P&L on \$1M	**+\$6,634** (+0.66%)
Best trade	SM 2021-Q3 (+24.45% / +\$24,453)
Worst trade	MTDR 2020-Q3 (–21.51% / –\$21,509)
Per-trade Sharpe (quarterly)	0.021
Max drawdown	–47.06%

Per-year:

Year	Cells	Trades	W/L	Hit	P&L on \$1M
2019	24	2	2/0	100.0%	+\$5,846
2020	28	3	2/1	66.7%	-\$464
2021	30	16	10/6	62.5%	+\$4,794
2022	38	9	5/4	55.6%	+\$10,210
2023	40	9	4/5	44.4%	+\$12,562
2024	40	12	4/8	33.3%	-\$26,313
Total	**200**	**51**	**27/24**	**52.9%**	**+\$6,634**

9.2 Distribution of returns

The distribution is diffuse rather than outlier-dominated. Top gains: SM 2021-Q3 (+\$24,453), OVV 2023-Q2 (+\$15,504), SM 2022-Q3 (+\$12,605), EOG 2022-Q3 (+\$10,605). Top losses: MTDR 2020-Q3 (–\$21,509), MTDR/OVV 2021-Q2 (–\$13,400 each), EOG 2022-Q4 (–\$10,059). No single trade exceeds 2.5% of capital. Excluding the single best leaves 50 trades at -\$17,819; the two best, 49 at -\$33,323 — moderately resilient to single-trade exclusion but not beyond, typical of a small-sample result. The 2020-Q1 cohort contributes one trade (OVV 2020-Q1, +\$16,900); SM 2020-Q1 does not fire (2019 trailing baseline of 9.5 active pads puts the forecast 0.5% *below* consensus, `in_line`).

9.3 Primary test (H1)

Pre-registered one-sided 5%-level firm-clustered bootstrap vs H0 = 50%. Observed **0.529**; **p = 0.328** (exact-binomial 0.390); 95% CI **[39.5%, 65.1%]**. **Fails to reject H0.** 95% CI on mean return [-1.83%, +1.53%] straddles zero. Quarter-block bootstrap yields [33.3%, 67.8%]. Directional estimate sits modestly above coin-flip but does not support a sharp claim of beating it.

9.4 Secondary test (H2)

H2 vs Strategy 3 (analyst-revision follower) was run on 2024 only; restricting S1 to 2024 trades (n = 12, 33.3%, **-\$26,313**) vs S3 (27.8% over 18 trades) — observed gap +5.5 pp exceeds the pre-registered +3 pp threshold on hit rate. Caveat: S1 exceeds S3 on hit rate but its 2024 P&L is more negative; H2 is technically supported on the metric but is not a positive-return claim — both lost money in 2024.

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10. Results: Cross-Strategy Comparison

10.1 Per-strategy hit-rate ranking — 2024

For the **Q1 2024 – Q4 2024** real-data sub-window, all signals are computed from real data (EIA WTI, EIA-DPR rig count, Sentinel-1 SAR, IBES consensus, CRSP+Yahoo prices). Strategies 3–10 are baselines whose pre-earnings hit rate is documented in the empirical asset-pricing literature.

Rank	Strategy	Hit rate	n_trades
1	S7 — XLE buy-and-hold	100%	1
2	**S1 — full multi-agent**	**33.3%**	**12**
3	S9 — value composite	33.3%	15
4	S6 — equal-weight universe	32.5%	40
5	S4 — WTI 3-month momentum	30.0%	20
6	S5 — Permian rig count	30.0%	10
7	S8 — 12-1 momentum	28.6%	14
8	S3 — analyst-revision follower	27.8%	18
9	S10 — quality composite	18.8%	16
—	S2 — no-news ablation	not run	—

All classical single-signal baselines earned 2024 hit rates between 19% and 33% — well below coin flip — reflecting 2024's regime (rangebound WTI, elevated sector volatility, anti-correlation between pre-earnings drift and several conventional signals). S7 is a single buy-and-hold. S1 fired 12 trades at 33.3% — tied with S9 for second among non-passive strategies, ahead of S3 by +5.5 pp (the H2 comparison in §9.4) — but S1's 2024 P&L was **-\$26,313**, the worst year. Outperforming a poorly-performing baseline on hit rate is not equivalent to making money.

10.2 Risk and drawdown

Small samples make Sharpe noisy; we do not run formal Sharpe-difference tests. S1's 2024 drawdown is driven by eight losing entries; in-universe baseline drawdowns are deeper (S4: –41%; S5: –10%; S6: –62%; S10: –47%) due to always-long exposure. S1's gating partially reduces drawdown but still fires enough losers to contribute.

10.3 Scope of claim

Descriptive ranking; pre-registered tests are in §§9 and 11. In 2024 every single-signal baseline lost money or barely broke even; S1 lost less per trade than several but more in aggregate because it fired more trades.

11. Ablations and Robustness

Pre-registered ablations are mechanically guaranteed by the Agent-3 gate; diagnostics and sensitivities follow.

11.1 RQ2 — $\alpha = 0$ ablation

$\alpha = 0$ makes Agent 2's forecast equal IBES consensus exactly; Agent 3 classifies every cell as `in_line`; gate short-circuits. **Result: zero long trades.** Agents 4 and 5 never fire, so the $\alpha = 0.10$ weighting is the system's only firing mechanism. Rules out the failure mode where Agents 4–5 generate spurious longs from qualitative inputs alone.

11.2 RQ3 — no-satellite ablation

Forcing Agent 1 to all-idle drives `drilling_signal` to zero. **Result: zero long trades.** Any non-trivial hit rate at the headline is a statement about the satellite signal *in conjunction with* the LLM scaffolding, not the scaffolding alone.

11.3 Agent-3 deterministic gate

Agent 3 computes class, percentage, and confidence deterministically in Python after the LLM call (text only); a schema validator rejects any output where the LLM's class disagrees with the rule. This is what makes §§11.1–11.2 mechanically guaranteed.

11.4 Defensive fallbacks and provider routing

Agent 4 has a permissive default on JSON/runtime errors. LLM calls route through OpenAI (`gpt-4o-mini` for Agents 2/3/4 and Bull/Bear; `gpt-5-mini` for Arbiter); sweep cost ~\$1–\$2 with caching. The deterministic core is provider-independent. SAR threshold sensitivity deferred.

11.5 Revenue-mechanism diagnostic

Joining Agent-3 class with Compustat `saleq` actual revenue: **41 of 51 long trades (80.4%) corresponded to actual revenue beats**; 22 of those 41 became stock wins (53.7%). The 10 cells with directionally-wrong forecasts produced 5W / 5L. A correctly-forecast revenue beat is necessary but not sufficient for a positive 2-trading-day exit — the reaction integrates EPS, capex guidance, hedge-book commentary, and the oil-price regime. Does not change H1.

11.6 WTI stress-veto sensitivity

Pre-registered point-in-time veto: block long entries with `wti_4w_return(T) ≤ −10.0%`. Sizing, costs, exit unchanged.

Veto blocks one trade — OVV 2020-Q1 (4w WTI = −16.8%, baseline +\$16,900). Post-veto: 50 trades, 52.0% hit, **-\$10,266** (vs +\$6,634 baseline); bootstrap p = 0.384, CI [38.1%, 64.8%]. The pre-registered veto is **not protective**: the −10% threshold is misaligned with the system's signal direction in the 2020-Q1 dislocation — OVV 2020-Q1 opened *after* the negative-pricing event and exited as oil partially recovered, capturing a regime-conditional mean-reversion the veto would have foreclosed. We do not retune. A single-threshold WTI guardrail removes a gain, not a loss.

11.7 Summary

(i) Remove satellite (or $\alpha = 0$) → system mechanically stops trading. (ii) Agent-3 gate eliminates LLM boundary noise. (iii) Core is provider-independent. (iv) Revenue mechanism correct on 80% of traded cells; monetisation adds variance. (v) Pre-registered WTI veto removes a gain, shifting +\$6,634 → -\$10,266. The constellation argues for caution about the +\$6,634 headline; +0.18% mean is within transaction-cost noise.

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12. Limitations and Threats to Validity

12.1 SAR change-detection rule

The Sentinel-1 RTC ingestion and per-pad backscatter are real, but the change-detection rule is intentionally simple: fixed +1.5/+0.5 dB thresholds on quarterly mean VV, anchored to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020). No CV/DL, no polarimetric decomposition, no coherent change detection on SLC products. A U-Net or fine-tuned EO foundation model (e.g., NASA-IBM Prithvi) would catch finer signals and emit per-pad probabilities. The system's most consequential simplification.

12.2 Sample size, IBES proxy, LLM determinism

n = 51 / 200 cells; bootstrap p = 0.328; 95% CI on mean return [-1.83%, +1.53%] straddles zero.

IBES consensus uses `ANNDATS_ACT` as proxy for pre-T-14 provability (Wall Street Horizon not subscribed). U.S. E&P firms pre-announce calendars 30+ days ahead — unbiased in expectation but subtly biased when announcement dates shift. The survivorship leak (Anderson & Akbas 2020) is mitigated by the `(REVDATS == ANNDATS) OR (REVDATS > T)` filter.

LLM determinism: temperature 0.0 + per-call cache. OpenAI does not guarantee bit-for-bit determinism but variance is small for constrained-JSON outputs; the deterministic core means trades are insensitive to small qualitative-text changes.

12.3 Universe, architecture, scope

The ten-firm universe selects on Permian concentration (correlated with mid-/small-cap status); choosing tickers known to exist as of 2025 is itself forward-looking. CRGY produces zero FracFocus Permian disclosures. 25-pad sample bandwidth-limited at ~12 hours wall-clock. Deterministic conviction-to-size lookup preserves traceability at the cost of bounded Sharpe. Strategies 2–10 are 2024-only; extending them is the most direct future-work item.

12.4 Recovery-regime over-firing and signal saturation

The most consequential structural limitation is **2021 over-firing**: 16 long trades in 2021 (post-COVID recovery) vs 2–12 in every other year. **13 of 16 (81%) reach `modest_beat` via a clipped `drilling_signal = +1.0`**, pinning divergence at +10.0%. Trailing-4Q active-pad averages in 2021 are dominated by the COVID 2020 trough (most pads idle), so baselines are low (1.25–7.75); 2021 active counts (7–23) divided by these produce raw signals of 1.4–7.5, all clipped.

The clip prevents runaway forecasts but the gate cannot distinguish modest from explosive recovery when the baseline collapsed. Agent-3 loses rank-ordering within the cohort: 16 trades at 62.5% hit but only +0.50% mean. We do not modify the clip post-hoc; §13 lists regime-aware sizing as future work. Excluding the 2021 cohort leaves 35 trades at 48.6% / +\$1,840.

12.5 External validity

Regime. Generalising to future regime breaks is not supported by the historical backtest — the 2021 over-firing pattern is specific to the post-COVID setup. **Crowding.** If the satellite signal becomes commercially available at scale, the edge compresses within 12–36 months per the signal-decay literature.

12.6 Documented portfolio rules vs enforced behaviour

The runner manifest records `max_names: 8` as a documented parameter, but the cap is not enforced in the inference layer. In this sample max simultaneous longs is 8 (2021-Q3, all eight then-eligible firms), so the cap never binds and the headline is unaffected. §8 describes it as documented-but-not-binding.

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13. Conclusion

We documented a multi-agent trading system for Permian E&P in which satellite-derived drilling activity feeds a deterministic revenue forecast assessed by an LLM investment board. Financial logic runs in code; qualitative outlook synthesis, news verification, and Bull/Bear/Arbiter debate run in language models.

This decomposition addresses the principal failure mode of agentic-AI trading: delegating unverifiable numerical decisions to language models, evaluated ex-post by P&L rather than ex-ante calibration. Our system avoids it by construction — every long trade traces to explicit numerical inputs; each LLM agent's marginal contribution is measurable by ablation. The $\alpha = 0$ and no-satellite ablations both produce zero long trades.

Empirical results on the real-data 2019–2024 window: **51 long trades**, 27W / 24L (52.9%), +0.18% mean net return per trade, median +0.88%, total **+\$6,634 on \$1M** (+0.66% of capital). Firm-clustered bootstrap fails to reject H0 = 50% (p = 0.328, 95% CI on hit rate [39.5%, 65.1%]); 95% CI on mean return [-1.83%, +1.53%] straddles zero. No single trade dominates; largest contributor is SM 2021-Q3 (+\$24,453). The pre-registered WTI stress veto blocks one trade (OVV 2020-Q1, +\$16,900) and shifts the aggregate to -\$10,266 — a single-threshold macro guardrail removes a gain, not a loss, in this window.

The most consequential structural finding is **2021 over-firing** (§12.4): 16 long trades in 2021, 13 clipped at `drilling_signal = +1.0` with identical +10.0% divergence. The clip works as designed but collapses signal differentiation across the recovery cohort. We disclose rather than tune.

We do not claim profitability after realistic execution costs (the +0.18% mean is within the 30-bps round-trip band), statistical sufficiency at n = 51, optimality of the SAR rule, or that the IBES `ANNDATS_ACT` proxy is fully look-ahead-clean. The contribution is methodological — a multi-agent design pattern with deterministic core, end-to-end real-data ingestion, and ablation-validated component contribution — portable to other alt-data-meets-fundamentals settings.

Four future-work avenues: (1) upgrade SAR change detection to a CV/DL pipeline; (2) regime-aware sizing/gating that conditions on trailing-baseline magnitude, restoring rank-ordering in recovery cohorts; (3) extend Strategies 2–10 to the full 2019–2024 window; (4) implement the documented 8-name concurrency cap or drop it.

The result is statistically modest: the system produces a directionally-positive signal whose marginal contribution is mechanically measurable, but the aggregate is small and not significant against the coin-flip null. The contribution is a discipline of separating deterministic from LLM roles, pre-registering tests, and running ablations that could falsify the central thesis — most important when an LLM is in the pipeline and when the result is unflattering.

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Appendix

A. Per-trade ledger, Strategy 1, 2019–2024

Long entries in chronological order; net return after 30 bps round-trip cost. Entry/earnings/exit dates and prices are in `runs/inference/strategy01_trades.csv`.

#	Cell	Size	Tier	Net return	W/L
1	SM 2019-Q2	0.05	low	+2.27%	W
2	DVN 2019-Q3	0.05	low	+9.42%	W
3	OVV 2020-Q1	0.10	medium	+16.90%	W
4	MTDR 2020-Q2	0.10	medium	+4.14%	W
5	**MTDR 2020-Q3**	0.10	medium	**-21.51%**	L
6	OVV 2021-Q1	0.10	medium	+2.72%	W
7	SM 2021-Q1	0.10	medium	-5.47%	L
8	FANG 2021-Q1	0.10	medium	+6.72%	W
9	OVV 2021-Q2	0.10	medium	-13.72%	L
10	MTDR 2021-Q2	0.10	medium	-13.36%	L
11	SM 2021-Q2	0.10	medium	-8.63%	L
12	FANG 2021-Q2	0.10	medium	+4.11%	W
13	OXY 2021-Q2	0.10	medium	+2.93%	W
14	MTDR 2021-Q3	0.10	medium	-0.47%	L
15	**SM 2021-Q3**	0.10	medium	**+24.45%**	W
16	FANG 2021-Q3	0.10	medium	+2.03%	W
17	DVN 2021-Q3	0.10	medium	+4.81%	W
18	OVV 2021-Q3	0.10	medium	-9.29%	L
19	CTRA 2021-Q3	0.10	medium	+0.49%	W
20	EOG 2021-Q3	0.05	low	+6.40%	W

21	OXY 2021-Q3	0.10	medium	+4.24%	W
22	DVN 2022-Q3	0.10	medium	+1.08%	W
23	EOG 2022-Q3	0.10	medium	+10.61%	W
24	SM 2022-Q3	0.10	medium	+12.61%	W
25	FANG 2022-Q3	0.05	low	+2.50%	W
26	OXY 2022-Q3	0.10	medium	-1.16%	L
27	OVV 2022-Q3	0.10	medium	+5.41%	W
28	DVN 2022-Q4	0.05	low	-12.19%	L
29	EOG 2022-Q4	0.10	medium	-10.06%	L
30	OVV 2022-Q4	0.05	low	-6.85%	L
31	SM 2023-Q1	0.10	medium	-9.73%	L
32	CTRA 2023-Q1	0.10	medium	-3.04%	L
33	MTDR 2023-Q2	0.10	medium	-1.41%	L
34	OVV 2023-Q2	0.10	medium	+15.50%	W
35	FANG 2023-Q2	0.10	medium	+8.06%	W
36	SM 2023-Q2	0.10	medium	+9.97%	W
37	FANG 2023-Q3	0.10	medium	-5.96%	L
38	CTRA 2023-Q3	0.10	medium	-5.65%	L
39	OXY 2023-Q4	0.10	medium	+4.82%	W
40	DVN 2024-Q1	0.10	medium	-2.49%	L
41	OVV 2024-Q2	0.10	medium	-8.51%	L
42	FANG 2024-Q2	0.10	medium	-6.80%	L
43	PR 2024-Q2	0.10	medium	-7.58%	L
44	SM 2024-Q2	0.05	low	-3.42%	L
45	MTDR 2024-Q3	0.10	medium	-1.75%	L
46	CTRA 2024-Q3	0.10	medium	-4.45%	L

47	SM 2024-Q3	0.10	medium	-6.23%	L
48	PR 2024-Q3	0.10	medium	+7.43%	W
49	OVV 2024-Q3	0.05	low	+6.93%	W
50	OXY 2024-Q3	0.10	medium	+0.88%	W
51	OVV 2024-Q4	0.10	medium	+1.43%	W

B. Signal-confidence score specification

A 0–100 deterministic score composing five 0–20 sub-scores from upstream-agent outputs: (1) SAR activity strength (`share_active`, binned by 0.0/0.15/0.30/0.45/0.60); (2) SAR newness (`n_newly_active / max(n_active, 1)`); (3) QoQ activity delta (`relative_activity_delta`, by sign/magnitude); (4) consensus tightness (inverse dispersion coefficient); (5) analyst panel breadth (binned by 3/6/10/15+). Tier mapping: `≥ 70 high`, `40–70 medium`, `< 40 low`. WTI is excluded so the score and the WTI veto (§11.6) do not double-count. Logged at `runs/inference/signal_confidence.csv`; does not enter H1. The 2021 over-firing pattern (§12.4) — 13 of 16 longs at identical clipped +1.0 signal — is a regime the score's sub-scores cannot rank-order.

C. Software, data, and computational resources

- **IBES Detail-History / Compustat / CRSP**: WRDS subscription.
- **yfinance Adj Close**: open-source, extends CRSP through 2025.
- **EIA weekly WTI spot** and **EIA-DPR Permian rig count**: EIA public release.
- **FracFocus public bulk CSV**: https://fracfocusdata.org/digitaldownload/FracFocusCSV.zip.
- **Microsoft Planetary Computer Sentinel-1 RTC**: https://planetarycomputer.microsoft.com (free, no auth).
- **GDELT 2.0 DOC API**: https://www.gdeltproject.org/.
- **LLM provider**: OpenAI API — `gpt-4o-mini` for Agents 2/3/4 and Bull/Bear; `gpt-5-mini` for the Arbiter.

D. Reproducibility

Every run writes `manifest.json` with SHA256 of input CSVs, SAR-mode flag, thresholds (1.5/0.5 dB), trailing-baseline coefficient (0.3), α, per-agent provider/model identifiers, prompt SHAs, Python version, and platform. LLM-call cache keyed on `(prompt_sha, input_sha, model_id, model_version, temperature)`. To reproduce: set `FIN580_SAR_MODE=real_sentinel1`, `FIN580_SAR_PADS_PER_OP=25`, execute Strategy 1 over the six yearly windows 2019Q1–2024Q4, then run `python -m fin580.inference.build_evidence_pack`.